

ET AI HACKATHON 2026

Zero-Harm

AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

Problem Statement 1 — Industrial Intelligence / Worker Safety / Geospatial Safety Analytics

Technical Documentation & Solution Architecture

Working Prototype · Architecture · Evaluation Methodology

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1. Executive Summary

Zero-Harm is a working prototype of a compound risk detection platform for heavy industrial plants. It directly answers the brief's core diagnosis: Indian industrial facilities are not short of sensors, permits, or maintenance systems — they are short of an intelligence layer that fuses those independent data sources into a single, real-time risk picture and acts on it before an incident, not after.

The prototype ingests four independent data streams — gas sensor readings, active work permits, maintenance activity, and worker location — and correlates them using a rule-based fusion engine to detect dangerous combinations no single system would flag alone. The flagship example, modelled directly on the January 2025 Visakhapatnam Steel Plant coke oven incident described in the brief, is the co-occurrence of a hot-work permit and rising gas levels in the same zone during active maintenance.

The deliverable includes a FastAPI backend implementing the Compound Risk Detection Engine and a live geospatial dashboard (the Geospatial Safety Heatmap deliverable) that visualises zone-level risk in real time, lists explainable alerts with recommended actions, and lets a judge trigger the incident scenario on demand to see detection happen live.

2. Problem Context

India's heavy industrial sector records thousands of fatal workplace accidents annually, and DGFIASLI figures cited in the brief put FY2023 fatalities above 6,500 — excluding most mining and construction. The January 2025 Visakhapatnam Steel Plant coke oven battery explosion is the reference incident for this problem statement: gas pressure sensors detected the danger, but no intelligence layer connected those readings to an operational decision in time, despite the facility having functioning gas detectors, permit-to-work controls, and SCADA.

A 2024 FICCI survey referenced in the brief found that over 60% of large industrial facilities still coordinate between their own digital safety tools through manual handoffs. The technology exists in silos; what is missing is fusion, correlation, and pre-emptive action — exactly the gap this prototype is built to close.

3. Solution Overview

The prototype implements two of the six illustrative components from the problem brief end-to-end, and is architected so the remaining four can be added as independent services without changing the core fusion logic:

Implemented in this prototype

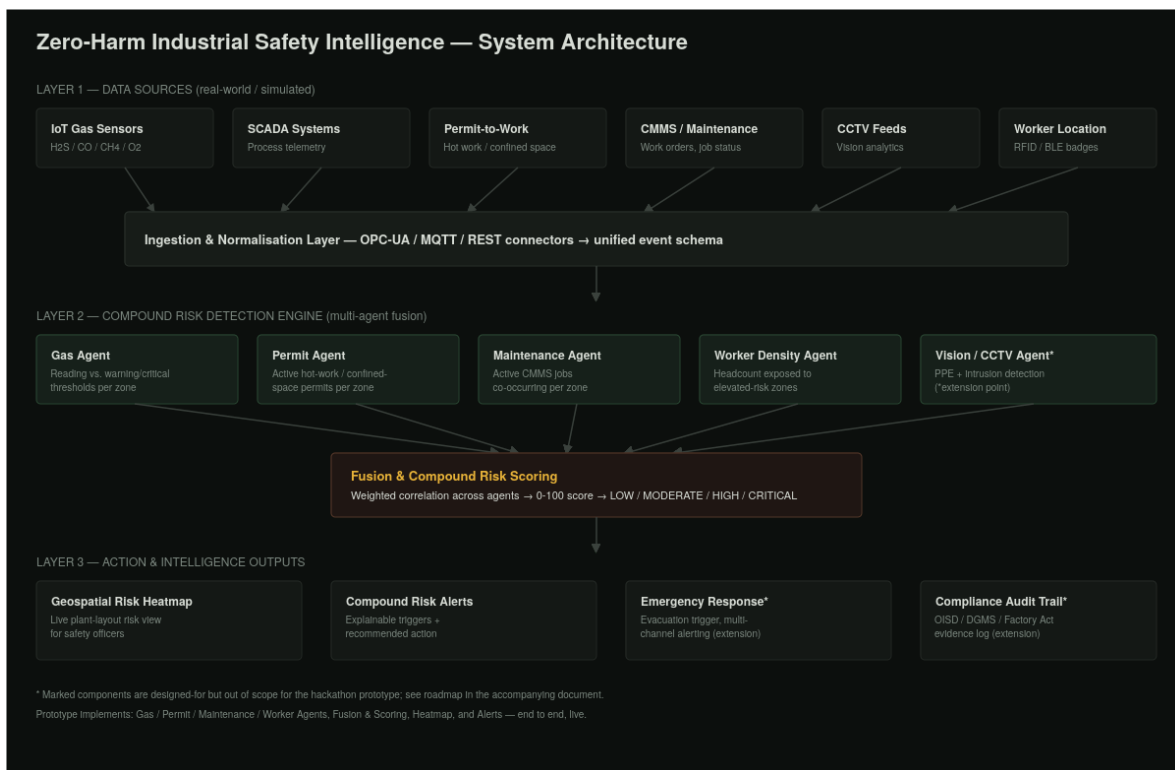
- Compound Risk Detection Engine — multi-source correlation across gas sensors, permits, maintenance activity, and worker location, producing a weighted 0–100 risk score per zone with explainable triggers.
- Geospatial Safety Heatmap — live plant-layout visualisation showing risk state per zone, updating every 3 seconds, with zone-level drill-down.

Designed for, extension points in the architecture

- Digital Permit Intelligence Agent — the permit/gas correlation logic already implemented is the core of this component; extending it to cross-check simultaneous operations across all permit types is a configuration change, not a redesign.
- Incident Pattern Intelligence — a RAG layer over historical near-miss and OISD/Factory Act guidance can subscribe to the same alert stream and add regulatory context.
- Emergency Response Orchestrator — the alert payload already contains a recommended action; wiring it to evacuation/notification channels is an integration task against the existing /api/alerts endpoint.
- Quality & Compliance Audit Agent — the permit and maintenance data model already carries the fields (zone, status, timestamps) an audit/compliance agent would need to cross-reference against OISD/DGMS/Factory Act requirements.

4. System Architecture

The platform follows a four-layer pipeline: heterogeneous data sources, an ingestion/normalisation layer, a multi-agent fusion engine, and action/intelligence outputs. Components marked with an asterisk in the diagram are designed-for extension points; everything else is implemented and running in the prototype.



In production, Layer 1 sources would connect via OPC-UA/MQTT (SCADA, gas sensors), REST/DB connectors (permit-to-work and CMMS systems), and RFID/BLE/UWB feeds (worker location) rather than the in-memory simulator used for the hackathon demo. No other layer needs to change to go live — the simulator was built to match the exact data shapes a real integration would produce.

5. Core Components

5.1 Data Simulator (data_simulator.py)

Models a six-zone plant layout (coke oven battery, gas cleaning plant, blast furnace bay, tar tank farm, sinter plant, rolling mill) matching the hazard classes referenced in the brief. Generates realistic gas sensor drift, seeds active permits and maintenance jobs, and exposes a `trigger_incident()` function that reproduces the compound failure pattern from the reference incident on demand.

5.2 Compound Risk Detection Engine (risk_engine.py)

Implements the fusion logic as four independent 'agent' functions — one per data source — whose outputs are combined by a single `assess_zone()` orchestration function. This mirrors a multi-agent pattern while staying simple enough to audit and explain to a judge in the demo: each agent can be swapped for a trained ML model later without touching the fusion or scoring logic.

Agent	Inspects	Detects
Gas Agent	Live sensor readings vs. per-gas warning/critical thresholds	Elevated or critical gas concentration by zone
Permit Agent	Active permits by type and zone	Hot-work or confined-space permits active during a hazard condition
Maintenance Agent	Active CMMS/maintenance jobs by zone	Maintenance activity co-occurring with a gas rise
Worker Density Agent	Worker badge locations by zone	Headcount exposed to an elevated-risk zone

5.3 API Layer (main.py)

A FastAPI service exposing the fused plant state and live alerts as REST endpoints, plus a background task that advances the simulation every two seconds so the dashboard feels live. Full endpoint reference is in Section 9.

5.4 Geospatial Safety Heatmap (frontend/index.html)

A SCADA-styled single-page dashboard: a zone map coloured by live risk level, an explainable alert feed (each alert lists the specific triggers and a recommended action, not just a score), and permit/sensor tables. Polls the API every 3 seconds; includes a zone selector and 'Simulate Compound Incident' control so judges can trigger the detection scenario live during a demo.

6. Compound Risk Scoring Methodology

Each contributing signal adds a calibrated weight to a zone's compound risk score (capped at 100). The weights below are illustrative starting points, consistent with the brief's evaluation note that thresholds should ultimately be tuned against historical incident data:

Signal	Weight	Rationale
Gas reading — WARNING	+25	Early indicator, not yet dangerous alone
Gas reading — CRITICAL	+45	Immediate hazard on its own
Hot work permit + elevated gas	+30	Classic ignition risk combination — the Vizag incident pattern
Confined space entry + elevated gas	+30	Asphyxiation / toxic exposure risk
Maintenance activity + elevated gas	+20	Unplanned intervention during a hazard window
≥2 overlapping permits in one zone	+10	Coordination risk between concurrent crews
≥3 workers present + elevated gas	+10	Exposure magnitude

Score bands map to action levels: 0–19 LOW (routine monitoring), 20–44 MODERATE (increase monitoring, notify supervisor), 45–69 HIGH (suspend relevant permits, notify safety officer), 70–100 CRITICAL (evacuate zone, suspend all active permits, dispatch emergency response) — directly operationalising the brief's requirement to 'trigger preemptive interventions before they escalate.'

7. Data Model

Core entities, implemented as typed Pydantic models so the API contract is enforced end to end:

Entity	Key fields	Source system in production
Zone	zone_id, name, x/y coordinates, hazard_class	Plant layout / GIS master data
GasReading	sensor_id, zone_id, gas_type, value_ppm, thresholds	IoT gas sensors / SCADA historian
Permit	permit_id, permit_type, zone_id, issued_to, validity, status	Digital permit-to-work system
MaintenanceActivity	activity_id, zone_id, description, active flag	CMMS (e.g. SAP PM, Maximo)
WorkerLocation	worker_id, name, zone_id, timestamp	RFID / BLE / UWB badge system
RiskAlert	zone_id, risk_level, risk_score, triggers[], recommended_action	Generated by the fusion engine

8. Technology Stack

Layer	Technology	Notes
Backend API	Python 3.12, FastAPI, Pydantic v2	Async, typed, auto-generates OpenAPI docs at /docs
Simulation	In-memory Python state + asyncio background task	Zero external dependencies for the hackathon demo
Frontend	Vanilla HTML/CSS/JS, SVG for the geospatial map	No build step — runs from static files served by FastAPI
Server	Uvicorn (ASGI)	Single command to run: uvicorn main:app
Suggested production additions	OPC-UA/MQTT client, PostgreSQL + PostGIS, a graph DB for equipment–permit–risk relationships, RAG stack (vector DB + LLM) for the Incident Pattern Intelligence extension	Not required for the current prototype

9. API Reference

Endpoint	Method	Description
/api/dashboard	GET	Single fused payload: zones, sensors, permits, maintenance, workers, alerts
/api/zones	GET	Plant zone master data
/api/sensors	GET	Current gas sensor readings
/api/permits	GET	Active and historical permits
/api/maintenance	GET	Maintenance/CMMS activity
/api/workers	GET	Current worker locations
/api/alerts	GET	Current compound risk alerts, highest score first
/api/simulate/incident/{zone_id}	POST	Triggers the demo compound-risk scenario in a zone
/api/simulate/clear/{zone_id}	POST	Resets a zone to baseline

10. Running the Prototype & Demo Script

Setup

1. `cd backend && pip install -r requirements.txt`
2. `uvicorn main:app --reload --port 8000`
3. Open `http://localhost:8000` in a browser

3-minute demo script (aligned to the required demo video)

4. Show the live dashboard at baseline — all zones green, sensor table nominal, explain the plant layout maps to a real coke-oven-battery-style facility.
5. Select 'Coke Oven Battery 3' and click 'Simulate Compound Incident' — narrate that this reproduces the Vizag pattern: gas levels begin climbing, an emergency maintenance job starts, and a hot-work permit is issued in the same zone.
6. Point out the zone marker turning amber then red on the heatmap within seconds, and the new alert appearing in the feed with its specific triggers listed (not just a score) and a recommended action.
7. Open the permits and sensor tables to show the underlying data the engine correlated — demonstrating this is a genuine fusion across independently-sourced signals, not a single-sensor threshold.
8. Click 'Clear Incident' to reset, and close by describing the four extension points in Section 3 that would take this from hackathon prototype to pilot-ready platform.

11. Evaluation Focus — How the Prototype Addresses It

Evaluation Focus (from brief)	How the prototype addresses it
Compound risk detection accuracy vs. single-sensor baselines	The engine only escalates to HIGH/CRITICAL when ≥ 2 independent signals co-occur (e.g. gas + permit); a single elevated sensor alone caps at MODERATE — directly demonstrable by comparing the sensor table to the alert feed.
Prediction lead time before incident threshold	Gas readings are shown trending through WARNING before reaching CRITICAL, with the alert already raised at WARNING when combined with a permit/maintenance trigger — giving lead time before the gas alone would have triggered a critical alarm.
Geospatial evidence quality	The heatmap renders every zone's real-time state with drill-down to the specific triggers, on a layout representative of the plant described in the brief.
Regulatory compliance coverage (OISD/Factory Act/DGMS)	Data model already carries the permit-type and zone fields a compliance-mapping extension (Section 3) would consume; full RAG-based compliance checking is scoped as a roadmap item (Section 13).
Reduction in false negative rate	Because the engine correlates across sources rather than relying on any single threshold, conditions that a single-sensor system would miss (e.g. a hot-work permit issued without any single sensor yet at CRITICAL) are still surfaced at HIGH.

12. Judging Criteria Alignment

Criterion	Weight	How this submission addresses it
Innovation	25%	Multi-agent compound correlation rather than single-sensor alerting; explainable triggers instead of an opaque score.
Business Impact	25%	Directly modelled on a real, named incident (Vizag, Jan 2025) and the FICCI-reported systemic gap of manual tool handoffs across 60%+ of large facilities.
Technical Excellence	20%	Clean separation of ingestion, fusion, and presentation layers; typed data contracts; swap-in path from simulated to real data sources without redesign.
Scalability	15%	Stateless FastAPI service; each agent and each zone assessment is independent, so horizontal scaling and additional zones/sensors require no architectural change.
User Experience	15%	Live, glanceable geospatial view for a safety officer, with explainable, action-oriented alerts rather than raw data dumps.

13. Roadmap to Production

- Replace the in-memory simulator with real OPC-UA/MQTT sensor feeds and a permit-to-work / CMMS API integration — no change needed to risk_engine.py's interface.
- Add the Incident Pattern Intelligence agent: a RAG layer over historical near-miss reports and OISD/Factory Act guidance, surfaced alongside each alert.
- Add the Emergency Response Orchestrator: wire /api/alerts to multi-channel notification (SMS/radio/PA) and an automated preliminary incident report generator.
- Replace rule-based agent thresholds with trained anomaly-detection models per gas type and zone, calibrated against historical incident data, while keeping the same fusion contract.
- Add a knowledge graph (equipment–permit–risk relationships) to support the Quality & Compliance Audit Agent described in the brief.
- Add authentication, audit logging of every alert and operator action, and integration with the plant's existing SCADA HMI for a single pane of glass.

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